K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

School of Engineering and Technology

Data Structures(ETCCDS202)

Assignment - 2

Assignment Title: Linear Data Structures Toolkit (LDST)

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Course Title : Data Structures

1. Introduction

This assignment focuses on the implementation and understanding of Linear Data Structures, where elements are arranged sequentially in a linear order. These structures form the foundation of efficient data storage and manipulation in computer science.

The objective of this project is to design and implement a Linear Data Structures Toolkit (LDST) using Python. The toolkit includes Dynamic Array, Linked Lists, Stack, Queue, and a Balanced Parentheses Checker.

2. Objectives

- To implement dynamic array with resizing
- To implement singly and doubly linked lists
- To build stack and queue using linked list
- To understand time complexity of operations
- To apply stack in solving balanced parentheses problem

3. Dynamic Array

Concept

A dynamic array automatically resizes itself when its capacity is full.

Working Principle

- Starts with a small capacity
- When full, capacity doubles
- Elements are copied into a new array

Amortized Time Complexity

- Most operations → O(1)

- Resizing → O(n)
- Overall → **Amortized O(1)**

Output

```
DYNAMIC ARRAY
Resizing from 2 to 4
Resizing from 4 to 8
Resizing from 8 to 16
1 2 3 4 5 6 7 8 9 10
Pop: 10
Pop: 9
1 2 3 4 5 6 7 8
```

4. Singly Linked List

Concept

Each node contains data and a pointer to the next node.

Operations

- Insert at beginning → O(1)
- Insert at end → O(n)
- Delete by value → O(n)
- Traversal → O(n)

Output

```
20 10 30 40
20 10 40
```

5. Doubly Linked List

Concept

Each node contains data, next pointer, and previous pointer.

Operations

- Insert after node → O(n)
- Delete at position → O(n)

Output

1 2 5 3
1 5 3

6. Stack (LIFO)

Concept

Stack follows Last In First Out principle.

Operations

- Push → O(1)
- Pop → O(1)
- Peek → O(1)

Output

Peek: 20
Pop: 20

7. Queue (FIFO)

Concept

Queue follows First In First Out principle.

Operations

- Enqueue → O(1)
- Dequeue → O(1)
- Front → O(1)

Output

Front: 10
Dequeue: 10

8. Balanced Parentheses Checker

Concept

Stack is used to check whether an expression has balanced parentheses.

Rules

- Opening brackets are pushed into stack
- Closing brackets must match the top
- Stack should be empty at the end

Test Cases

([]) → Balanced
([)] → Not Balanced
(((→ Not Balanced
"" → Balanced

9. Output (Execution Result)

DYNAMIC ARRAY
Resizing from 2 to 4
Resizing from 4 to 8
Resizing from 8 to 16
1 2 3 4 5 6 7 8 9 10

Pop: 10
Pop: 9
1 2 3 4 5 6 7 8

SINGLY LINKED LIST
20 10 30 40
20 10 40

DOUBLY LINKED LIST
1 2 5 3
1 5 3

STACK
Peek: 20
Pop: 20

QUEUE
Front: 10
Dequeue: 10

PARENTHESES CHECK
([]) → Balanced
([)] → Not Balanced
(((→ Not Balanced
"" → Balanced

10. Conclusion

This project helped in understanding how linear data structures work in practical scenarios. Dynamic arrays handle resizing efficiently, linked lists allow flexible operations, and stack and queue are important data structures.

These data structures are widely used in real-world applications such as browsers, compilers, and memory management systems.

11. References

- GeeksforGeeks
- Python Documentation
- Class Notes

12. Execution Note

The program was successfully executed using Python, and the output was generated through the terminal and stored in output.txt.